# Carryover Monitor Executive Summary

Team THD | Rockwell Automation and NVIDIA Physical AI Hackathon | CSI Fluid Mixing Station 3

Team THD developed an advisory carryover monitor for shared-path fluid filling. The system combines recipe context with real-time vial imagery to warn operators before a risky recipe transition and to check whether the filled vial shows visible carryover. Our prototype demonstrates that this approach can run in real time on an NVIDIA DGX Spark, while also identifying the additional real-line validation and integration needed before production use.

## Manufacturing problem and goal

Several source tanks feed a shared dispensing path. When the recipe changes, residual liquid can enter the first vial of the next batch and alter its color or composition. A missed defect may cause scrap, rework, added inspection, downstream quality problems, or an unnecessary line interruption. Our goal was to give operators early, explainable evidence while keeping the monitor read-only and advisory.

## Physical AI approach

The process model compares the current and upcoming recipes to identify transitions with carryover risk. A compact neural network evaluates the vial's developing color from camera frames, while interpretable color rules provide confirmation evidence. The DGX Spark dashboard shows the active recipe, upcoming queue, live frame, model probability, and event history. A rolling three-of-five-frame rule reduces reactions to isolated noisy frames. The monitor recommends inspection, diversion, or an approved flush; it does not change PLC recipes or operate the line.

## Data and use of CSI equipment

We used CSI digital-twin assets and a reproducible pipeline to create 30 fps vial-fill sequences at approximately live filling speed. Runs varied recipes, liquid colors, expected hue, carryover severity, and imaging conditions, with synchronized video, queue, and frame-level metadata. We also evaluated recorded footage from the real CSI vial-fill camera in Dataset 5. Because the supplied timing labels drifted away from the visible fills, we detected fill periods from the images and created corrected alignment labels. The Spark executed the frozen model and dashboard replay at original timing. We did not complete an untouched live production trial: the PLC-controlled machine was unavailable during the first visit, and authorized access to the line-connected Spark and camera was unavailable during the second.

## Measured evidence

| Scored vials | Carryover detected | False alarms | ROC AUC | Mean inference |
|---|---|---|---|---|
| **23** | **6 of 7** | **2** | **0.839** | **4.1 ms** |

At the balanced evaluation threshold of 0.0265, the model detected six of seven expected carryover cases, missed one, and produced two false alarms, for 85.7% recall, 87.5% specificity, 75% precision, and approximately 87% accuracy. The dashboard uses a lower 0.0071 threshold to reduce missed defects, so it intentionally produces more inspection warnings and should not be assigned the balanced-threshold accuracy figures above.

## Manufacturing value and deployment path

Earlier warning and visual confirmation could reduce defect escapes, manual inspection, rework, and troubleshooting time. Recipe-aware monitoring may also support targeted flushing instead of flushing after every change, reducing material waste and downtime. The concept applies to paint, chemical, pharmaceutical, food and beverage, and personal-care processes that share dispensing paths. Production deployment would add an approved read-only PLC or MES recipe feed, authorized live-camera access, more labeled real fills, threshold calibration on separate validation data, and a final untouched test set.

Supporting material https://github.com/ALL-Applied-AI-Network/msoe-ai-club-rockwell-automation-nvidia-hackath-thd-16c769df76